
Talk with your latent - can LLM do scientific reasoning directly from data?

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Abstract

Recent advances in Large Language Models (LLMs) offer unique opportunities for scientific tasks, yet their ability to reason over complex numerical data remains largely unexplored. While several works investigate the ability of LLMs to create textual descriptions of scientific objects (e.g., proteins), we ask whether they can understand those objects and answer comparative questions about them. This kind of reasoning beyond description is crucial for the development of useful co-scientists. We test two different astrophysical datasets, involving both stars and galaxies, with pre-trained multimodal features that serve as the LLM data tokens. Adding those tokens as additional information, we fine-tune an 8B Llama model on various scientific reasoning tasks. We find that while the model can describe the stellar objects reasonably well, they fail when asked about populations, outlier detections, and cross-modal correlations. We further show that by training a small non-language head that would highlight the scientific information (e.g., a flow matching model for population questions), and adding its latent features as an extra token, the performance of the LLM on the relevant tasks improves dramatically. The results are consistent across two independent datasets and highlight the lack of statistical reasoning abilities in current language-based models and the need for dedicated models to transform the data in a way that helps LLMs understand it.

1 Introduction

The integration of Large Language Models (LLMs) into the scientific workflow marks a paradigm shift in how researchers interface with information. While these models have demonstrated remarkable proficiency in some scientific tasks, their utility remains largely confined to the textual and visual domains. A critical frontier in “AI for Science” is the transition from processing scientific text to reasoning directly over the physical data that defines scientific discovery. Standard LLM architectures struggle with numerical reasoning [1], in part because subword tokenization does not preserve the magnitude and continuity of numerical values [2], creating a gap between natural language and empirical data.

The main motivation of this work is the need to combine physics foundation models and LLMs in a scientifically useful manner. Each has distinct advantages, and the potential for knowledge alignment between them has far-reaching outcomes. One is interpretability: an LLM that operates over both natural language and learned scientific representations could serve as a ‘bridge’ between humans and machine learning representations, which are usually hard to interpret. A second is scientific discovery: foundation models trained on large and diverse datasets encapsulate latent information beyond their downstream tasks, and that information can support non-trivial population analysis or complex cross-modal correlations. Unfolding it, however, requires analysing high-dimensional features directly, whereas the common practice of projecting them to a low-dimensional summary

discards exactly the structure such analyses depend on. Most current uses of LLMs with scientific data take the form of textual descriptions, or of agents that delegate the analysis to dedicated models and read off their results. Whether an LLM can perform scientific reasoning *directly* from learned data representations is therefore still an open question, and it is the question we study here.

Contribution. We connect frozen survey encoders to an 8B language model through a trained projector, adapting the language model itself with either LoRA or full fine-tuning, and evaluate it on two independent astrophysical datasets (Kepler-field stars, COSMOS-Web galaxies) across per-object and pairwise questions, and on a relational question for each: the odd member of a set of stars, and the agreement between two views of a single galaxy. *First*, we find a sharp asymmetry: reading learned tokens, the model describes a single object and compares two objects accurately on both datasets, but fails on the scientific reasoning tasks — identifying the anomalous member of a group of stars, and recognising when a galaxy’s morphology and star-formation history disagree — where it performs at or below chance. A probe on the underlying representation recovers most of that signal in both cases, so the failure is one of computation rather than representation; and for the stellar task, supplying the same catalog values as text in the prompt leaves the model at chance once its verbosity is corrected for, while prompting it to reason through the objects first lifts it off chance but not to the level the tokens already reach. *Second*, we show that performing the set-level computation outside the model and returning it as a single extra token recovers the ability: +0.33 on outlier identification over the full test fold, and +0.47 on an adversarial split where the odd member was never an odd member during training. Scrambling or zeroing the token’s contents costs significantly on every axis but coverage. *Third*, the head must match the question. A density-based head recovers rarity among stars but is at chance on cross-modal agreement among galaxies, where a discriminative cross-modal predictor works instead. *Finally*, we report two evaluation traps that each would have produced a confidently wrong result, and find that verified templated prose yields a substantially more factual model than fluent LLM-written prose on the same task.

2 Related Work

LLMs as scientific assistants. Advances in LLMs have already had a wide-ranging impact on how we do science. LLM agents can now review and write scientific papers [3, 4], act as a ‘virtual lab’ that suggests and implements novel research ideas [5], or perform equation discovery [6]. However, these agents typically access scientific data indirectly, by writing code to an external tool (e.g., ESM, AlphaFold-multimer), an approach formalised in the tool-augmentation literature [7, 8]. Recently there has been growing interest in predicting physical properties of proteins with LLMs [9, 10]. That line is close to ours, but asks a different question: property prediction and textual description from sequential data, rather than comparative and population analysis over continuous data.

Scientific foundation models and scientific LLMs. Foundation models have become popular across scientific fields and show strong performance in many domains [e.g., 11–18]. Notably, these models do not treat natural language as a modality, so a clear separation persists between scientific foundation models and LLMs. The term scientific-LLM usually refers instead to injecting scientific knowledge into existing LLMs [for reviews, see 19, 20], most often by fine-tuning on high-quality text and images to improve reasoning [21–23], object description [24], search [25], and chat [26]. Such models outperform general LLMs on domain tasks [27], but remain confined to text and images.

Connecting frozen encoders to frozen language models. The architecture we use is standard outside the sciences: a trained projector maps a frozen encoder’s outputs into the embedding space of a frozen language model, which is adapted with a small number of trainable parameters [28–31]. This recipe is well-established for producing descriptions of a single input. Whether it supports comparison across several inputs, which is what population-level scientific questions require, has received little attention.

3 Data and Model

We test two different astrophysical datasets, representing different domains and physical properties. The first dataset is a *stellar* multimodal dataset comprising 190,091 stars, observed by 5 different

modalities coming from different telescopes - *Kepler*[32], TESS[33], LAMOST[34], APOGEE[35], and Gaia[36]. Each modality is processed through a specific encoder, and the modalities are aligned with a model that factorizes the latent space into shared and specific subspaces. The result is one shared feature and 5 per-modality features, each projected to the token embedding space of the LLM as its own small group of tokens. The second dataset is galaxies from the COSMOS-Web survey [37], including two modalities: images and star-formation history (SFH). Similar to the stellar dataset, we use a pretrained model that aligns the two modalities (here using CLIP [38]) and project the frozen features into the LLM’s token embedding. Notably, the two datasets are different in many aspects: number and type of modalities, pretrained models, and scientific questions. Using the two datasets in the same pipeline highlights general vs data-specific conclusions.

training prose. One of the challenges in this pipeline is creating the training dataset. This should include ground-truth prose per object. Contrary to other fields, in astrophysics data is usually stored only in numerical catalogs, so one needs to create the prose from scratch. Our main method is a *template-based* method: a deterministic program that picks a subset of facts from the catalog, chooses a phrasing from a bank of templates, and writes prose. Every sentence is checked against the record by a verifier before it enters the corpus. The check is per-fact: it guarantees that the facts a description states are true of the object, not that the description mentions every unusual object in a group. Both datasets share two types of prose: description of a single object (single), and comparison between pairs (pair). We also test deeper reasoning capabilities with two additional tasks. In the stellar dataset we use population prose: a group of N objects with description of the typical members, and a callout of the one that does not belong and on which property. In the galaxy dataset we are interested in non trivial cross modal correlation and check if the model highlight those in the prose. In Section 4 we compare the template-based method with a second method: an *LLM-based* generation, where we let DeepSeekV4 write the population prose from catalog facts, and verify their correctness. **Input tokens.** To improve the diversity of the data and reduce overfitting, we apply several augmentations to the input. First, we add a random number of sentences from general astrophysics papers (from arXiv) and insert them around the data tokens. Second, we add label smoothing and a cosine learning-rate schedule. These augmentations apply to the single and pair records; population records are presented as the data tokens alone.

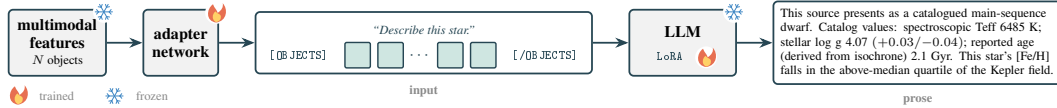


Figure 1: **The pipeline.** An adapter projects each object’s multimodal features into tokens placed between brackets; the model continues from them into prose, and the loss is taken on the prose alone. The language model is adapted either by full fine-tuning or with LoRA.

4 Results

4.1 singles and pair questions

We first test the single and pair questions, comparing two types of Llama-8B models: Llama-3.1 and AstroSage [27], a Llama-3.1 model that was fine-tuned specifically on arXiv astrophysics papers. Table 1 shows the comparison on both datasets. While both models show high performance on both tasks, Astrosage retains a small gain over the vanilla model in seven of the eight cells, across both datasets, which implies that textual domain knowledge is helpful for those tasks. Therefore, the Astrosage model was chosen for further analysis.

4.2 scientific reasoning tasks

Single and pair questions consists of description and simple comparisons and requires almost no reasoning beyond the basic properties. It is also similar to the protein description tasks as in [10], and therefore it is not surprising that an LLM performs well on such questions. Next, we test the performance on questions that requires some form of scientific reasoning. As mentioned in Section 3, those are anomaly detection for the stellar dataset and irregular morphology-SFH correlation in the galaxy dataset. Table 2 shows the results of those two tasks using both a template-based verifier

and an LLM judge (Qwen3.5-27B). The *no head* column presents the basic model, with no extra information, and the performance is poor. Before adding anything to the model, we check whether this is a failure of information. A cross-fit linear probe on the same aligned features, asked to rank the members of a group, matches what the true catalog values achieve (rank-1 0.660 from the shared subspace and 0.686 from the full feature, against 0.660 for the catalog itself; Section A). The signal the task needs is present in the features the model reads, so the failure is in the readout and not in the representation. The *+ head* column shows the performance with the addition of an extra single token that comes from a dedicated non-LLM model that is meant to highlight the relevant information. In the stellar case this is a flow matching model that was trained to learn the joint distribution of the aligned features and catalog labels, conditioned on the rest of the group. In the galaxy case, this is a cross-modal predictor that is trained to predict one modality from the other. We note that adding this single token significantly improves the performance on both tasks.

Table 1: **Base model: Llama-3.1-8B vs. AstroSage-8B**, identical corpus, recipe and seed; rule-scored, greedy; the better model in bold. Metrics: *numeric acc.* = fraction of numbers the text states that fall within tolerance of the catalog value; *qualitative acc.* (single) = fraction of class words (dwarf/giant, spiral/elliptical, ...) that match the catalog; *relational acc.* (pair) = fraction of comparisons (“hotter than”, “more massive”) that match; *outlier found* = fraction of groups — among those that contain an odd member — in which the text names the true outlier *and* its property. [†] Population rows come from an earlier 150-group subsample and are comparable only to each other, not to the full-fold population results.

metric	Llama-3.1-8B	AstroSage-8B
<i>galaxies — one object</i>		
numeric acc.	0.822	0.826
qualitative acc.	0.812	0.835
<i>galaxies — pair</i>		
numeric acc.	0.647	0.672
relational acc.	0.795	0.771
<i>stars — one object</i>		
numeric acc.	0.909	0.911
qualitative acc.	0.575	0.628
<i>stars — pair</i>		
numeric acc.	0.784	0.801
relational acc.	0.858	0.883

Table 2: **The head fixes the relational task — on both tracks, under both scorers.** Each row is one arm pair on one task, measured two ways. Stars: does the text name the true odd member *and* its property, over the 1,203 test groups that contain one. Galaxies: does the text call the shape-vs-history relation correctly, over the 1,484 galaxies where the two genuinely *disagree*. The rule verifier and the LLM judge are independent instruments; they agree on the size and direction of the stellar effect. The galaxy rows are scored differently, and * explains why; their two rows are two controls (a narrator trained without the token, and the same trained narrator with its token removed at test time), not two scorers. Numbers are comparable *within* a row block only — the two tracks are different tasks on different scales.

track — relational task	scorer	no head	+ head	Δ
<i>stars</i> — name the odd member and its property	verifier	0.245	0.633	+0.388
	judge	0.332	0.662	+0.330
<i>galaxies</i> * — does shape agree with history?	min-stratum AUC	0.442	0.628	+0.186
	min-stratum AUC	0.461	0.628	+0.167

* The galaxies are scored by how well the model’s graded confidence separates true mismatches, taken as the minimum over the two star-formation strata (0.50 = chance), rather than by whether the text states the right verdict. A verdict rate saturates here: trained on the same prose, the narrator without the token calls 0.923 of the mismatching galaxies correctly and the one with it 0.973 — both say the right words. What separates them is that the narrator without the token learns “spheroid

⇒ mismatch”, which is right for the 90% star-forming majority and backwards for the quiescent minority.

4.3 is it the token, or the retraining that came with it?

The head arms differ from the no-head arms in more than the token: they are trained with LoRA on a frozen base, while the no-head arms are full fine-tunes. A reader is entitled to ask whether the gain is the token at all. The test that does not depend on the training regime is to take the *same trained model* and destroy the token at generation time, leaving everything else untouched. Table 3 does this two ways: replacing the token with zeros, and scrambling its contents across the group so that it carries the same statistics but no longer describes the star it is attached to.

Table 3: **Breaking the token at test time**, stellar group task, full test fold ($n=1,203$ groups with an odd member, greedy, judge-scored). *found* = the text names the true odd member and its property. The middle two rows are the *same* trained model as the last one, with its token damaged at generation time — same weights, same training regime, only the token changes. *p* is exact McNemar against the intact arm.

	found	<i>p</i> vs. intact
no head token at all	0.332	2×10^{-70}
+ head token, zeroed	0.561	2×10^{-11}
+ head token, scrambled	0.606	2×10^{-5}
+ head token, intact	0.662	—

What it says. Scrambling the token costs 0.056 and zeroing it costs 0.101, both significant. The token is not a flag whose mere presence helps: what it *contains* is read and used. We quote the scrambled row as the headline, because zeroing pushes the input off the distribution the model was trained on while scrambling does not. The galaxy track shows the same thing — removing the cross-modal token at test time drops the trained narrator from 0.628 to 0.461 (Table 2).

This settles the content of the token but not its presence: the gap between no token at all (0.332) and a zeroed token (0.561) still mixes the token being there with the LoRA regime that came with it. To separate them we trained the missing cell on the galaxy track — same prose, same LoRA regime, no token — where the cost is affordable. It reaches a min-stratum AUC of 0.498 ([0.462, 0.535], indistinguishable from chance), against 0.442 for the full fine-tuned no-token arm and 0.628 with the token. The training regime on its own buys nothing on the interaction task; both no-token arms sit at chance on the minority stratum and learn the morphology shortcut instead. The token, not the regime, is what carries the result. Section A gives the complete 2×2 .

4.4 is it reasoning, or is it recall?

The stellar prose corpus was not built split-aware: 99.2% of the stars in the test fold appear in *some* training description. That is symmetric across arms, so the paired contrasts above are fair, but it leaves an obvious objection — the head arm might be recalling which star was called unusual during training rather than working it out.

We test this directly, on the stellar track. From the test fold we keep only groups whose odd member was *never* an odd member anywhere in training: the star may have been described many times, but never as the one that stands out. Only 2.3% of candidate groups survive that filter, leaving 1,000. If the head is recalling, it should lose most of its advantage here.

What it says. The advantage does not shrink where memorisation is impossible — it grows. The head is worth +0.469 on this set against +0.330 on the full fold, and the two arms move in opposite directions when they are moved onto it: the arm without the head falls (0.332 → 0.279) while the arm with it does not. That is the opposite of what a recall explanation predicts. We wrote the prediction down before running it.

One caveat on reading the table. This set is deliberately harder than the full fold and its item mix is different, so only the within-set gap is quotable: compare 0.748 to 0.279, not to the 0.662 of Table 2. Groups with no odd member are excluded throughout, here and everywhere else in the paper.

Table 4: **The adversarial set**, 1,000 stellar groups whose odd member was never an odd member in training, greedy, judge-scored. *found* = the text names the true odd member and its property. *p* is exact McNemar (Wilcoxon for contradictions).

	found	named	property	covered	all correct	contra./group
latent tokens (baseline)	0.279	0.329	0.343	0.560	0.017	1.85
latent tokens + head	0.748	0.782	0.761	0.628	0.140	1.35
Δ	+0.469	+0.453	+0.418	+0.068	+0.123	-0.50
<i>p</i>	3×10^{-105}	8×10^{-104}	1×10^{-86}	1×10^{-5}	1×10^{-28}	3×10^{-25}

4.5 giving the model the numbers as text does not help

If the features carry the signal and the model cannot use it, an obvious alternative is to skip the features and hand the model the catalog numbers directly. We test this on the *stellar* dataset only, where the group task has a written form: the same base model is shown each group as an indexed table of numeric fields — more fields than the head ever sees — and asked, in words, to name the member that stands out and the property it differs on. We also test a version that is told to reason through the stars one at a time before committing to an answer. Table 5 shows both against the two latent arms on the same groups.

What it says. The text arm looks competitive on the raw rate (0.592) but it names 68% of the group and never stops generating, so the rate is mostly hedging: corrected for how much it names, it is below chance ($0.87\times$). On **found**, where it must also name the right property, it scores 0.331 against 0.320 for the latent tokens — the two are statistically indistinguishable — and both are less than half the head’s 0.652. Reasoning first makes the arm commit (18% of the group named instead of 68%, and it terminates) but does not rescue it: 0.224, the lowest of any arm that finishes. Reading the same numbers as text is therefore not a route to the group task, with or without explicit reasoning.

Table 5: **The catalog-text control**, the $n=884$ groups with an odd member judged in all four arms, greedy. **found** = outlier and property, the paper’s consistent metric. *outlier named* is the index alone and *lift* divides it by the fraction of the group the arm names (1.0 = chance given that much hedging): the two together show *why* the index alone misleads. Under **found** the text arm’s apparent lead is gone — 0.331 against 0.320 for latent tokens, both less than half the head’s 0.652. *stars named* and *finished* describe all 1,500 generations, not the judged subset.

	found	outlier named	lift	stars named	contra./group	finished
catalog table, text	0.331	0.592	$0.87\times$	68% of group	6.74	0/1500
catalog table, text, reason first	0.224	0.397	$2.21\times$	18%	2.58	1249/1500
latent tokens	0.320	0.431	$3.32\times$	13%	1.88	1498/1500
latent tokens + head	0.652	0.782	$3.91\times$	20%	1.58	1500/1500

4.6 the head has to match the question

The two heads are not interchangeable, and we know because we tried. Our first attempt on galaxies was to port the stellar head unchanged: a density model over the aligned features, scoring each object by how unlikely it is. On stars that is the right question — the odd member of a group really is a rare object. On galaxies it found nothing.

Table 6 shows why we abandoned it. For each axis we compare the density head reading the full galaxy features against the *same* head given only the catalog labels. If the features carried usable signal for this head, the first column should beat the second. On eight of the nine axes it is worse, including the cross-modal axis this paper cares about (0.589 against 0.662). The features are not adding anything the head can use.

What it says. This is not a broken model or a useless representation: the *same* galaxy features, read by a cross-modal predictor instead of a density model, support the task at 0.628 (Table 2). What changed is the question the head asks. A density model scores *rarity* — how unlikely is this object. The galaxy question is *disagreement* — does the image agree with the star-formation history. A star-forming spheroid is a perfectly ordinary galaxy that happens to sit on the wrong side of a boundary,

Table 6: **The wrong head on galaxies.** AUC for ranking the odd member of a group (400 populations, test fold), for a density head reading the full aligned features vs. the same head given only catalog labels. Negative gain means the features make it worse.

axis	features	labels only	gain
morphology–SFH mismatch	0.589	0.662	−0.073
Sérsic shape	0.477	0.915	−0.438
quenching	0.588	0.902	−0.314
redshift	0.462	0.725	−0.263
age	0.435	0.624	−0.189
mass	0.687	0.844	−0.157
SFH trend	0.567	0.680	−0.113
cluster member	0.508	0.557	−0.049
disturbed	0.544	0.527	+0.017

so it is not rare, and a likelihood cannot see it however well it is trained. The stellar anomalies, by contrast, genuinely are rare.

So the recipe is not “attach a head”. It is “attach a head that computes the quantity the question turns on”. Deciding which one that is — rarity, or agreement between two views — is a modelling choice made before any training, and getting it wrong costs everything: the same pipeline, the same data and the same narrator produce a result at chance.

4.7 verified templates beat fluent LLM prose

Everything above is trained on prose written by the deterministic templater. The obvious alternative is to let a capable LLM write it, which reads far better and is much less work. We built that corpus too: DeepSeek-V4 was given the same catalog facts and asked to write the group descriptions, and every claim it made was checked against the record before the text was accepted. The training text is, in that sense, true. Table 7 compares the two corpora with everything else held fixed — same base model, same prompt, no head.

Table 7: **Which prose to train on.** Same base model, same prompt, no head; the 1,191 groups with an odd member judged in both arms, paired, greedy. Coverage, contradictions and all-correct differ at $p < 10^{-45}$; *outlier found* at $p = 9 \times 10^{-6}$.

training prose	outlier found	covered	contradictions/group	all correct
template (ours)	0.333	0.573	1.88	0.128
DeepSeek-V4 (verified)	0.258	0.178	7.16	0.000

What it says. Training on fluent LLM prose makes the model invent: contradictions per group go from 1.9 to 7.2, and not one group out of 1,191 is described fully correctly. The training text was verified true; the model trained on it is not.

The two corpora differ in how much they commit to. Per group description, the templater writes a median of 21 words containing 5 numbers and names 17% of the members — it states the population and the member that does not belong, and says nothing about the rest. The teacher writes 82 words containing 17 numbers and names 83% of the members, giving most of them a full quantitative profile. Training on the second teaches the model to produce that profile for every object in every group. At generation time on a group it has not seen, the tokens do not support reading off five exact values per object, so it supplies them anyway. Verification of the training text does not help here, because the problem is not that the text is false — it is that it asserts more than the input determines.

5 Limitations

The corpus is not split-aware. 99.2% of test-fold stars appear in some training description. The contrast is symmetric across arms so the paired comparisons hold, but absolute rates are optimistic, and the adversarial set (Table 4) is the answer to the recall objection rather than a clean test. A split-aware rebuild is the right fix and we have not done it.

248 **The ground truth names one intruder when several qualify.** The composer designates one off-
249 axis member and leaves the others unconstrained on that axis, so 60.5% of anomalies sit in a group
250 containing at least one other member with the same label (81.7% on dwarf/giant). Naming any
251 of them is a true statement, but only one is scored correct. Re-scoring both arms to credit any
252 qualifying member raises both and leaves the head’s lead intact ($+0.361 \rightarrow +0.318$), so this does
253 not drive the result — but every absolute rate in this paper is a lower bound for that reason. The
254 same construction makes the “no anomaly” groups unreliable: 71.4% of them contain a member
255 that crosses a threshold on some axis the composer did not check, which is why groups with no
256 designated odd member are excluded from every number we report.

257 **One seed per arm.** Every cell was trained once. Differences of a few points between individual
258 cells — particularly in Table 1 — are within the noise we cannot measure. The claims rest on the
259 pattern across cells and on effects large enough that a seed is not a plausible explanation.

260 **The regime control exists only on galaxies.** Table 9 closes the LoRA-vs-full-fine-tune confound
261 on the galaxy track. The stellar equivalent would cost ~ 700 – $1,000$ GPU-hours and was not run, so
262 the stellar claim inherits that control by analogy rather than directly.

263 **We can show the token is used, not how.** The ablations establish that the token’s contents are
264 read (Table 3), but not the mechanism. Two things remain unexplained. When the head’s own top
265 pick is wrong, the narrator does not preferentially follow it, so it is not simply reading the head’s
266 answer off the token. And most of the token’s channels are unstable under re-sampling — eight of
267 eleven have a run-to-run spread comparable to their spread across objects — so the information the
268 narrator uses is probably carried by the few channels that are stable, not by the token as a whole. We
269 report the effect; we do not claim to have explained it.

270 **The judge is an LLM, not a human.** Population records are scored by a 27B model. It agrees
271 closely with the independent rule-based verifier, which is the evidence we have that it is not inventing
272 its own standard, but there is no human-labelled subset. A few hundred hand-scored groups would
273 settle it.

274 **The two tracks do not carry the same evidence.** The catalog-text control and the adversarial set
275 exist only for stars; galaxies have no set-level task at all, and their relational question is cross-modal
276 agreement within one object rather than comparison across a group. Where a claim rests on one
277 track we have said so.

278 **No claim about scale.** A 70B narrator was trained on galaxies and is excluded here: it was not
279 a controlled comparison against the 8B arms (batch 16 vs 128 at the same unscaled learning rate),
280 and it degenerates into repetition on long outputs. We therefore say nothing about how these results
281 move with model size.

282 6 Conclusion

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594 A Appendix

595 A.1 Is the signal in the features? The ranking probe

596 The group task asks which member does not belong. We test whether that is answerable from the
597 aligned features alone, without a language model in the loop. For each group we rank its members
598 by one quantity and ask where a qualifying star lands. Two readouts are compared on exactly the
599 same groups: the true catalog value, and a cross-fit ridge regression that predicts the same quantity
600 from the aligned feature (5-fold, so no star is scored by a model that saw it). If the probe tracks the
601 catalog, the signal survives the alignment.

602 Two details matter. First, only the three continuous axes can be ranked by a scalar; binarity and
603 cluster membership are categorical and are excluded (103 of 259 anomalies). Second, the composer

designates one intruder but leaves the other members unconstrained on that axis, so a group often contains several stars carrying the same label — 60.5% of anomalies on the full fold, and 81.7% on dwarf/giant. Naming any of them is a true statement about the group, so a readout is credited for the best-placed qualifying star rather than only the designated one. “Extreme” is distance from the group median in units of its median absolute deviation, which does not reveal which direction the answer lies in.

Table 8: **Ranking the odd member from the features alone.** $n=156$ continuous-axis anomalies over 129 groups. *rank-1* = a qualifying star is the single most extreme member; *top-3* = one is among the three most extreme. The probe reads only what the language model reads.

readout	rank-1	top-3
catalog values (oracle)	0.660	0.942
probe, shared subspace (64-d)	0.660	0.840
probe, full feature (1024-d)	0.686	0.891
<i>by axis, full feature vs. catalog</i>		
dwarf / giant	0.911 vs 0.857	1.000 vs 0.929
hot / cool	0.755 vs 0.694	0.939 vs 0.959
metal rich / poor	0.373 vs 0.412	0.725 vs 0.941

The probe matches the catalog at rank-1 and keeps 89–95% of it at top-3; on the gravity axis it is ahead of the catalog. The information the group task needs is in the features.

The catalog row is not a ceiling on the task. It bounds what a *single ranked quantity* can do, and it falls short of 1.0 because the label is a threshold test (“[Fe/H] > 0.3”) while a ranking is a distance test: a star can sit far from the group median without crossing the threshold. A readout that models the group jointly — the head of Table 2 — is not bound by it.

Scoring the narrators by the same rule, crediting any equally-qualifying star, raises both arms and leaves the head’s lead intact: $+0.361 \rightarrow +0.318$ ($n=1,203$ groups, $p=6 \times 10^{-85}$).

A.2 The regime control: same prose, same LoRA, no token

The head arms are trained with LoRA on a frozen base while the no-head arms are full fine-tunes, so “head vs. no head” is also “LoRA vs. full fine-tune”. We closed that on the galaxy track, where one cell costs ~ 10 GPU-hours: a narrator trained on the *same* concordance prose, with the *same* LoRA rank and schedule as the head arm, and no cross-modal token. It is one variable away from the head arm.

Table 9: **The complete** 2×2 , galaxy cross-modal task, minimum-AUC over the two star-formation strata (0.50 = chance; brackets are 95% intervals on the minority stratum). A score above 0.5 in *both* strata detects the interaction; below it, the arm is reading morphology alone.

	full fine-tune	LoRA	read
no token	0.442 [0.409,0.477]	0.498 [0.462,0.535]	morphology-only
+ token	—	0.628 [0.593,0.665]	interaction

Neither training regime produces an interaction detector on its own: both no-token arms sit at chance on the quiescent stratum, and the LoRA arm’s interval contains 0.5. The full fine-tune is in fact the more confident shortcut learner — it posts the highest *marginal* AUC in the study (0.885) while scoring 0.442 on the minority stratum, which is why we never quote marginal AUC here. Only the arm with the token is above chance in both strata. We take this as evidence, by analogy, for the stellar track as well, where the equivalent control would cost ~ 700 – $1,000$ GPU-hours and was not run.